MILESTONES 1 – 5 | 427 TESTS PASSED | COMPLETE STACK REFERENCE

Smart Database Visualizer

Complete Architectural Evolution & Tech Stack Specification (Milestone 1 to Milestone 5)

Version: Milestone 5 Release
Stack: Next.js 16 + Express + MySQL
Status: 100% Fully Functional

1. Milestone Evolution Matrix (Summary Overview)

MILESTONE	PRIMARY SCOPE	CORE TECHNOLOGIES & LIBRARIES	KEY FILES & WORK AREAS
Milestone 1	Backend Foundation & Secure DB Connectivity	Node.js v24, Express 4.19, `mysql2/promise`, Zod, Helmet, CORS, Rate-Limit, Dotenv	`backend/src/server.js` `backend/src/config/database.js`
Milestone 2	Zero-Config Dynamic Schema Introspection	MySQL `INFORMATION_SCHEMA`, Parameterized SQL Queries, Normalization Parser	`backend/src/services/schemaService.js` `frontend/lib/schema.ts`
Milestone 3	Interactive ER Canvas & Hierarchical Auto-Layout	Next.js 16, React 19, TypeScript, `@xyflow/react` (React Flow), `dagre`, Tailwind CSS 4	`frontend/components/visualization/*` `frontend/lib/schema/dagreLayout.ts`
Milestone 4	Rule-Based Health Engine & Dependency DAG	Structural Graph Algorithms (DAG), `STATISTICS` Introspection, Recommendation Engine	`backend/src/services/analysis/*` `recommendationBuilder.js`
Milestone 5	Table Explorer, Read-Only SQL, CRUD & Stabilization	5-Tab Table Explorer, AST Read-Only Tokenizer, 13-Point FK Pre-Validator, Base UI / Radix	`TableDetailsPanel.tsx`, `queryService.js` `schemaEditService.js`

2. Milestone-by-Milestone Deep Breakdown

Milestone 1 — Backend Architecture & Connection Infrastructure

FOUNDATION

Constructed an enterprise-grade Express REST backend establishing connection pools to MySQL without credential leaks.

Node.js (v24 LTS) | Express (4.19.2) | mysql2/promise (3.10.1) | Zod (3.23.8) | Helmet (7.1.0) | CORS (2.8.5) | express-rate-limit (7.3.1) | Nodemon (3.1.4)

- **Connection Pool:** `createPool` with non-blocking async TCP sockets, connection testing, and automatic recovery.
- **Input Validation & Sanitization:** Zod schemas validating port ranges, connection payloads, and regex allowlist (`[a-zA-Z0-9_]`).
- **Security Middleware:** Helmet for 11 security headers, CORS origin-locking, and 500 req/15 min IP rate limiting.

Milestone 2 — Schema Introspection & Normalization Engine

INTROSPECTION

Eliminated hardcoding by introspecting live schema catalogs via parameterized SQL queries and standardizing JSON output.

MySQL INFORMATION_SCHEMA | Parameterized SQL | REST Controllers | TypeScript Models

- **Zero-Config Discovery:** Queries `TABLES` (base tables), `COLUMNS` (types, nullability, auto-increment), `KEY_COLUMN_USAGE` (foreign keys), and `TABLE_CONSTRAINTS` (primary/unique constraints).
- **Normalized Representation:** Maps dynamic MySQL metadata into `DatabaseSchema` (tables array, column lists, relationships array with N:1 cardinality).

Milestone 3 — Interactive Graph Visualization & Auto-Layout Engine

VISUALIZATION

Engineered an interactive, high-performance canvas rendering tables and relationships with automatic Dagre layout.

Next.js 16 (App Router) | React 19 | TypeScript 5.7 | @xyflow/react (React Flow 12.12) | Dagre (0.8.5) | Tailwind CSS 4 | Lucide React

- **Canvas System:** Pan, infinite zoom ($0.15\times - 2.5\times$), mini-map, and custom table cards (`TableNode.tsx`) with badges (`PK` , `FK` , `UQ` , `IDX`).
- **Custom Edge Renderer:** `RelationshipEdge.tsx` with SVG cubic bezier curves, target direction arrows, and interactive cardinality chips.
- **Size-Aware Dagre Engine:** Injects dynamic table heights ($\text{Height} = 42 + N \times 28\text{px}$) to compute collision-free Left-to-Right (`LR`) and Top-to-Bottom (`TB`) coordinates.

Milestone 4 — Rule-Based Health Inspection Engine & Dependency DAG

HEALTH & DIAGNOSTICS

Integrated a graph-theoretic health inspection engine executing 5 rules and calculating a project-defined health score.

Graph Algorithms (DAG) | STATISTICS Metadata | Rule Engine (R001 - R005) | Health Score Formulation

- **Rules Implemented:** R001 (Missing PK - HIGH), R002 (Orphaned Table - MEDIUM), R003 (Missing FK Candidate - MEDIUM), R004 (Unindexed Foreign Key - LOW), R005 (Circular / Dependency Anomalies).
- **Health Scoring Formula:** $\text{Score} = \max\Big(0, \, 100 - (15 \times \text{High} + 8 \times \text{Medium} + 3 \times \text{Low})\Big)$.
- **Structural Dependency Graph:** Categorizes tables into root nodes, leaf nodes, and isolated tables with full dependency chains.

Milestone 5 — Table Explorer, Query Console, CRUD & Full Stabilization

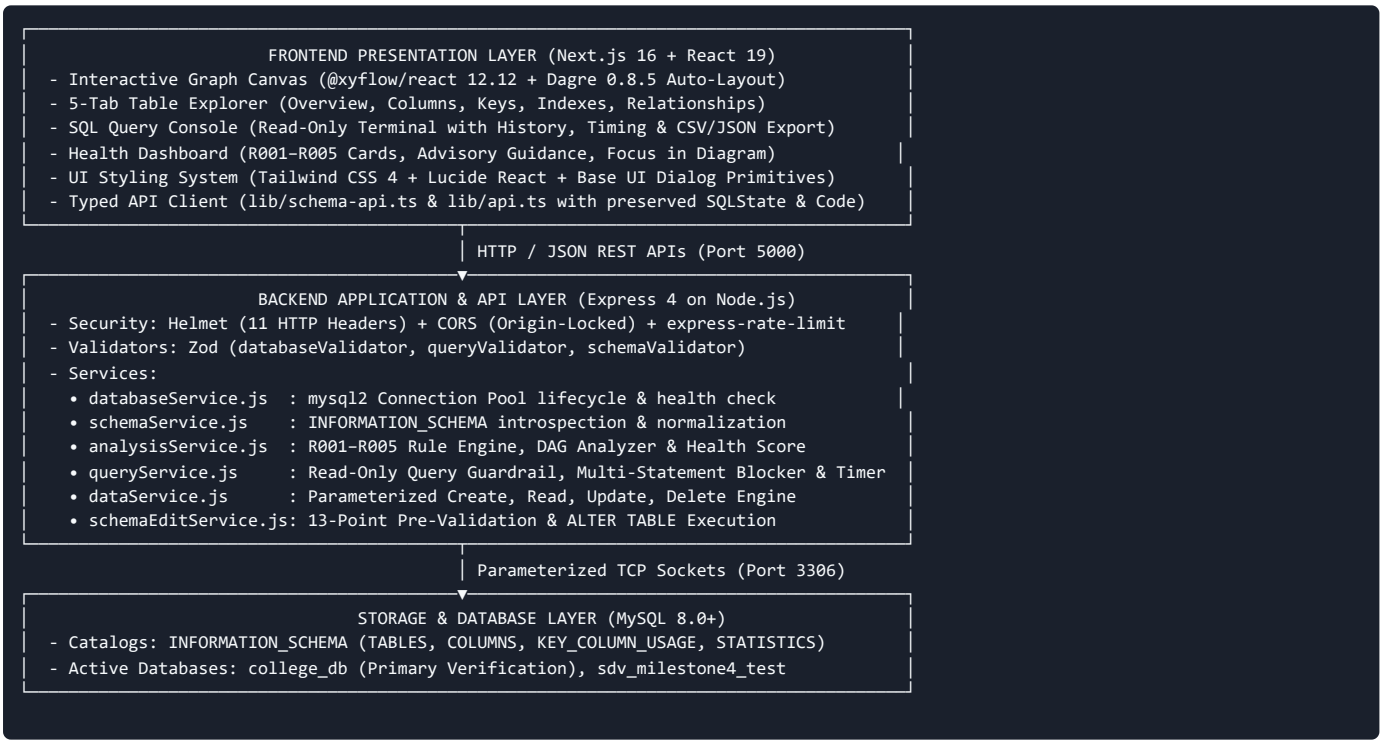
INTEGRATION & PRODUCTION

Unified all modules with in-depth table exploration, read-only SQL terminal, live CRUD, and 13-point FK pre-validation.

5-Tab Explorer | Read-Only SQL AST Parser | 13-Point FK Pre-Validator | Base UI / Radix | Parameterized CRUD | Local Visual Filters

- **5-Tab Table Explorer:** Detailed inspection of Overview, Columns, Keys & Constraints, Indexes (from `STATISTICS`), and Relationships.
- **Read-Only Query Console:** AST tokenizer allowing `SELECT/SHOW/DESCRIBE/EXPLAIN` and read-only `WITH`; blocks mutations with `READ_ONLY_QUERY_CONSOLE` and chained SQL with `MULTIPLE_STATEMENTS_NOT_ALLOWED`. Preserves MySQL error codes (`ER_NO_SUCH_TABLE`, `42S02`).
- **Pre-Validated FK Engine:** 13 checks (types, sign, target indexing, orphan records, constraint naming) before running `ALTER TABLE`.
- **Declared vs. Possible Separation:** Confirmed FKs render solid blue; R003 heuristic candidates render dashed amber and never mutate declared schema.

3. Complete Cumulative Layer Architecture



4. Automated Test Matrix Across All Milestones

TEST SUITE	MILESTONE SCOPE	VERIFICATION COVERAGE	STATUS
`backend/test-api.js`	Milestones 1 & 2	Express API routing, 404 handlers, Zod sanitization, credential hygiene	**26 / 26 PASS**
`backend/phase1-test.js`	Milestone 2	Schema extraction, catalog parsing, foreign key resolution	**55 / 55 PASS**
`backend/phase2-test.js`	Milestone 4	Rules R001 (Missing PK) and R002 (Potential Orphaned Table)	**49 / 49 PASS**
`backend/phase3-test.js`	Milestone 4	Rules R003 (Missing FK), R004 (Unindexed Column), and R005 (DAG Dependency)	**42 / 42 PASS**
`backend/phase4-test.js`	Milestone 4	Analysis summary aggregation and project-defined health score (47 on test DB)	**59 / 59 PASS**
`phase5-test.js`	Milestone 5	Database health dashboard integration and diagram focus verification	**51 / 51 PASS**
`phase6-test.js`	Milestone 5	SQL Console execution, SELECT/SHOW/DESCRIBE/EXPLAIN, Read-only rejection	**50 / 50 PASS**
`phase7-test.js`	Milestone 5	Full CRUD lifecycle (Create, Read, Update, Delete) on MySQL tables	**37 / 37 PASS**
`e2e-audit-test.js`	Milestone 5	Comprehensive end-to-end integration audit on `college_db`	**58 / 58 PASS**
TypeScript Check	All	Strict typing via `pnpm exec tsc --noEmit`	**0 ERRORS**
Production Build	All	Production asset bundling via `pnpm build` (Next.js 16)	**SUCCESS (3.0s)**